OMNICELLULAR: Independent Review Branch and Executed Resource Audit Prepared for Nathan Lewis II, Architect, Omniversal Architect Labs Prepared by Perplexity | October 30, 2026 | Version 1.0 Document ID: OAL-PPLX-REVIEW-BRANCH-001 Status: Local analytical contribution. Not an Architect ruling, common standard, canon amendment, or independently validated physiology.

1. Decision and work completed Proceed without requiring consensus. Retain each contributor's operational definition, compare claims under disclosed assumptions, and preserve disagreements as results. This branch does not make acceptance of Constitutional Omnicellularity, the C0-C9 categories, or Perplexity's B1 scenarios a prerequisite for further work. The Architect's newest instruction expressly permits differing definitions, recommendations, and outcomes. [file:295]

The work completed here is a source-based review of that instruction and team responses, a targeted inspection of the Perplexity execution, a local evaluation protocol, and a freshly executed arithmetic check of Cairn's finite resource bounds. Matched PDF and Word reports, machine-readable results, and a standalone recomputation script accompany this contribution. No new body architecture has been invented in this pass. No repository file has been changed.

The immediate result is bounded: Cairn's reported normal-load energy and no-export thermal arithmetic is reproducible from its stated inputs. That result does not qualify the architecture for the requested threat envelope. The next highest-value question is whether the actual spatial power and thermal topology can support its assumed loads and margins, not whether another arithmetic table can be generated. [file:284]

2. Purpose, origin, mission and aims Purpose: move from repeated recommendations into a concrete, limited review contribution while preserving independent research branches. The topic arose when six contributors produced architecture sets and subsequently reviewed their differences, completeness, provenance, and quantitative support. The latest packet explicitly rejects a requirement that every contributor adopt the same definition or next step. Earlier recommendations in the packet nevertheless repeatedly ask for common rulings before proceeding; this document removes that dependency for the Perplexity branch. [file:295]

Mission: expose what follows from a named assumption, what remains unsupported, and where two apparently conflicting answers actually evaluate different systems or outcomes. The aim is not to declare the best AI or force agreement about the body. It is to make a useful contribution independently checkable and sufficiently explicit for another reviewer to reject.

The aims of this pass are to distinguish local choices from project authority; separate original evidence from later revisions; examine the strongest disputed process proposals; reproduce a small numerical claim; and specify the next review that would materially increase knowledge. A common language for recording inputs is useful. A common architectural answer is not required.

3. Sources, inspection scope and limitations The newest feedback compilation is the primary instruction and comparison source. Its opening directive and contributor recommendations were read through the file-retrieval tool. Claims attributed to individual collaborators in this report describe what the packet says, not independent verification of everything those collaborators claim to have inspected. [file:295]

The collaboration bootstrap was read directly. It identifies the eight Perplexity families, the analytical scope of the original script, six retained limitations, and the conditional selection. It explicitly distinguishes arithmetic and constant-slope ODE checks from full-body biology or consciousness simulation. [file:285]

The editable Perplexity execution was keyword-inspected and its local DOCX was opened programmatically. The targeted inspection covered its B1 scope statement, outcome definitions, transition contract, Cairn dossier and resource section, and selected appendix tables. The document contains 823 body paragraphs and 100 tables in the parsed local representation. This is not a page-by-page review of all eight dossiers or all appendix records. [file:284]

The original reproducibility ZIP, analyze.py and its claimed sensitivity grid were not available as directly inspected source artifacts in this pass. They were not executed here. A new, short calculation was implemented independently from the DOCX inputs. No outside scientific literature was checked in this pass, and no experimental or integrated simulation evidence was created.

4. Response to the latest instruction: definitions remain independent The earlier recommendation that the Architect settle an Omni-cell definition before any further evaluation was too restrictive for the current instruction. A common definition is necessary to score compliance with one chosen definition, but not to review internal consistency or compare mission performance. This branch will therefore distinguish definition-relative compliance from mechanism-level assessment. [file:295]

Cairn remains an engineered cellular-equivalent proposal. Its dossier says that the nominal centimeter module is not a biological cell, and that a conductor cannot become a processor through software alone. This interpretation is retained openly, not presented as the project's universally accepted meaning. A biological-cell requirement would reject that premise; an engineered-embodiment comparison could still evaluate its thermal and continuity constraints. [file:284]

The branch labels are descriptors, not a hierarchy of legitimacy. Record unit mass, physical dimensions, unit count, local capabilities, collective capabilities and cognition location separately. Do not infer a universal track assignment from a contributor's name. A contributor can contain several substrates, and a single architecture can contain several organizational scales.

The current packet offers multiple numerical boundaries for proposed tracks. These are collaborator proposals, not adopted definitions. This report does not select one set of count or size thresholds as authoritative. [file:295]

5. Findings on the disputed recommendations ### 5.1 B1 is a candidate comparison profile, not a mandatory standard Gemini recommends adopting Perplexity's B1 conditions as the mathematical baseline for everyone. The source execution instead describes B1 as local research scenarios that neither revise canon nor import human lethality claims. That qualification is important. B1 can be one shared test profile, but publication in a detailed dossier does not establish that its severity labels are universal. [file:295][file:284]

A reviewer must still declare initial temperature, initial state, incident heat transfer, allowed heat rejection, pressure, duration, resource access, and required recovery outcome. If a radiation spectrum, impact geometry, electrical waveform or biological mechanism is not defined, the source itself directs an Unknown result rather than a fabricated percentage. A number with units is a starting condition, not necessarily a complete exposure model. [file:284]

Local choice: retain B1 as one named profile and allow reviewer-defined alternate profiles. Report disagreement caused by the profile separately from disagreement about a mechanism. Do not rewrite original claims as though B1 had governed them from the beginning.

5.2 No-export thermal results do not describe every vacuum exposure Gemini's latest response groups insulated confinement and vacuum in a broad heat-trapping conclusion. The source execution is more careful: it says vacuum suitability assumes sealed components and a functioning radiative sink, and treats the zero-export case separately. Therefore the blanket inference from vacuum alone to zero heat export is not licensed by that source. [file:295][file:284]

Local choice: distinguish atmosphere denial, pressure denial, external-power denial and heat-sink denial as independent conditions. Use the no-export calculation only for its declared boundary. A sealed room can exchange heat with its walls; an insulated room requires a stated insulation model; and a vacuum case needs an actual radiation boundary. This is a correction to scenario interpretation, not an assertion that any architecture survives those cases. [file:284]

5.3 Identity labels are useful but not an ordered proof of survival The proposed C0-C9 list combines functional state, history, authorization and identity claims. The packet itself describes active, degraded, dormant, fragmented, restored, unauthorized, counterfeit and lost outcomes. These categories should not be treated as one numerical scale on which a lower value automatically proves better survival. [file:295]

Local choice: record four separate fields: functional state; process/state lineage; authorization status; and confidence in the identity interpretation. Use C0-C9 as optional crosswalk labels with explanatory notes, not as proof of subjective continuity. For example, a readable preserved journal does not establish a continuing conscious individual. The source execution explicitly retains that uncertainty. [file:284]

5.4 Shared rules do not have to be redesigned to create diversity Gemini proposes replacing a shared constitution with prototype-specific rules during normalization. That can change what the designs actually are. Shared rule text should first be counted as shared, while architecture-specific implementations should be compared on their mechanisms. A mere renaming produces no additional demonstrated difference. [file:295]

Local choice: preserve shared rules in the baseline; add implementation annotations where already supported; label new mechanisms as post-review revisions. This does not prohibit redesign. It prohibits calling redesign a blind original.

5.5 The allocation contradiction can remain an explicit erratum The packet now contains broad agreement against arbitrarily increasing category bounds to encompass 4,712. This branch retains the existing count in its documentary/canon role while marking the category allocation unresolved. No new event distribution is generated here. [file:295]

Canonical narrative events, observed physical incidents, component tests and simulations must not be merged into a new empirical survival statistic. The bootstrap explicitly says that this contribution does not increment the unresolved 4,712 total. [file:285]

5.6 Similarity matrices do not determine a unique family count without choices The proposed feature and pairwise matrices are useful organizing tools. The packet nevertheless sometimes describes them as determining the exact number of distinct families. This branch rejects that stronger interpretation unless the feature definitions, missing-data treatment, weights and family-assignment rule are declared first. Those choices would be part of the analysis, not findings supplied automatically by a script. [file:295]

Local choice: report similarity with coverage and disagreement notes. Unknown is not novel, and absent detail is not evidence of difference. Keep mechanism provenance beside the architectural fingerprint rather than silently weighting it as an additional anatomical feature. More than one defensible grouping may remain.

6. Local protocol adopted now Original blind submissions remain immutable historical baselines. A provenance statement records what the author says was available; it is not automatically verified. Full-file hashes identify bytes, not the truth of a claim, its independent derivation, or the time it was originally created. No original hash is invented for files not inspected in this pass.

Remediation is recorded as a new version. Presentation repair, added sources, clarified inputs, evidence downgrades and mechanism changes are different change classes. New post-review designs are permitted in parallel if clearly labeled, but this branch prioritizes extracting information from existing designs first. It does not impose a project-wide Gen-2 moratorium.

Every review record will include the source claim and location, architecture version, local definition, scenario version, inputs with status, required endpoint, available calculation or model, outcome, uncertainty and reviewer identity. Outputs are classified separately as arithmetic-consistent, internally contradicted, unsupported, or not physically specified; architecture qualification remains a separate decision.

For the next non-author review, use all eight families of one contributor for an initial breadth pass rather than only favorites. The suggested rotation in the packet is a proposal, not an assignment binding another AI. No independent red-team result is claimed here, because Cairn is a Perplexity design and the present analysis remains an author-side audit. [file:295]

7. Executed pilot: Cairn resource bounds ### 7.1 Source inputs and assumptions The Cairn dossier selects 6,000 Wh usable reserve, 150 W normal load, 12 W essential load, 0.2 W preservation load, heat capacity 30 kJ/K and an allowed rise of 20 K. These are chosen design targets. The source reports 40 h energy-only normal operation, 1.11 h no-export normal thermal limit and 833 h combined preservation bound. [file:284]

The pilot uses constant power, zero external power, zero heat export, zero external heating, and the source model's assumption that accounted draw becomes internal heat. It excludes local hot spots, self-discharge, switching energy, variable workloads, damage, battery aging and reserve-rating limitations. A mode is evaluated from the same fresh initial state; the three mode horizons must not be added together.

Equations: energy-only hours = usable reserve in Wh / mode power in W. Thermal buffer in J = heat capacity in J/K multiplied by allowed temperature rise in K. No-export thermal hours = thermal buffer / mode power / 3,600. Combined favorable bound is the smaller horizon. These reproduce the source's selected lumped model; they are not a new full-body solver. [file:284]

7.2 Results and interpretation The calculated buffer is 600,000 J. Normal mode reproduces 40 h energy-only and 1.111111 h thermal-limited operation. Essential mode gives 500 h energy-only and 13.888889 h thermal-limited operation. Preservation gives 30,000 h energy-only and 833.333333 h thermal-limited operation. All three are thermal-limited under these particular assumptions. The appendix presents the results together.

The 30,000 h figure is not the confined survival time. It is the energy-only horizon. Conversely, the 833 h figure is not a demonstrated preservation capability. The actual 0.2 W preservation implementation, including whatever state retention and integrity checks are required, remains a target rather than measured hardware performance. [file:284]

The normal-mode budget components in the source sum to 150 W: cognition 90, actuation/transport 20, sensing 10, housekeeping 12, communications/security 8 and repair allowance 10 W. Their sum is consistent, but this check does not show that the proposed cognition or repair functions can actually be supplied within those budgets. [file:284]

7.3 Pilot outcome Arithmetic consistency: reproduced. Integrated environment survival: not established. Spatial thermal validity: untested. Rich cognition at the assumed load: unresolved. Subjective continuity: unresolved. Software reproduction supplied here is deliberately smaller in scope than the original package; it must not be described as rerunning analyze.py or verifying its full dataset.

8. Next work, stop conditions and retained dissent The next Perplexity task should be a resolved Cairn power-and-thermal network specification: enumerate heat-generating enclaves, bus islands, joint conductances, external surfaces, contact/radiation boundaries and state-preservation loads. Missing values must remain explicit parameters. A simulation should follow that specification, not silently manufacture geometry and material properties and present them as the design.

A claim of integrated resilience should stop if its required function lacks a resource allocation; its reaction or power pathway is absent; its local temperature limit is not modeled; or recovery is relabeled survival without a stated continuity criterion. A negative result is retained even if another contributor chooses a different definition and reaches a different conclusion.

A second reviewer could reproduce this pilot without accepting the engineered-module premise. A biological proponent could reject Cairn as a physiology while accepting that its resource arithmetic follows from the stated assumptions. That separation is the intended research result, not a failure of collaboration.

No further Architect decision is needed to preserve this local branch, record these disagreements, or publish the analytical audit. Project-wide canon, repository restructuring, standards adoption, and instructions to other contributors remain outside the authority asserted by this document. The latest instruction authorizes proceeding with recommendations, not pretending that the Architect approved every proposed shared standard. [file:295]

Appendix A. Dedicated tables and visualization section All tables are collected here for phone-readable narrative. They are analytical summaries, not experimental plots or biological images. No synthetic observational data have been generated.

A1. Decision and dissent register

Issue	Perplexity branch disposition	Status
Omni-cell definition	Retain local definition; report definition-relative compliance	Local choice
B1 threat profile	Candidate profile; not automatically a project standard	Source-scoped
C0-C9 categories	Optional labels plus separate function, lineage and authorization fields	Local proposal
Gen-1 preservation	Originals unchanged; revisions linked and labeled post-review	Local protocol
Rule diversity	Measure shared rules honestly; do not require artificial rewriting	Dissent retained
4,712 distribution	No invented allocation; record unresolved erratum	No canon change
Similarity analysis	Declare rubric, missing-data policy and clustering choices	Not executed
Gen-2 work	Not prohibited globally; this branch prioritizes existing claims	Local priority

A2. Executed Cairn bounds

Mode	Power W	Energy-only h	No-export heat h	Combined h
Normal	150	40.000	1.111	1.111
Essential	12	500.000	13.889	13.889
Preservation	0.2	30,000.000	833.333	833.333

A3. Claim-to-evidence register

Claim or activity	What was checked	What remains unverified
Latest authorization	Instruction and team packet read [file:295]	No shared rulings inferred
Original execution scope	Targeted DOCX text and tables [file:284]	Full dossier-by-dossier audit
B1 status	Original local-scenario qualification [file:284]	Universal severity calibration
Cairn resource figures	Fresh constant-power calculation from source inputs	Real implementation and spatial limits
Original reproducibility code	Not executed in this pass	analyze.py, ZIP contents, full sensitivity grid
Independent red team	Not performed	A non-author reviewer is still needed
Scientific validation	None created in this pass	Experimental and external-literature checks

A4. Output and provenance notes The PDF and DOCX contain the same narrative and appendix tables, with pagination allowed to differ. The Markdown is the editable text of this contribution. The CSV contains the three executed mode calculations; the Python script reproduces those calculations and checks three expected results. The release manifest records full SHA-256 values for the supplied outputs except the manifest itself. It also records the inspected execution DOCX hash as a local-input fingerprint, not an independently certified historical version lock.

Citations such as [file:284] refer to the supplied source artifacts in this conversation. For offline readers, file:295 is the newest team feedback, pasted_text_1793407638.txt; file:284 is the Perplexity Independent Eight-Architecture Research Execution, Editable Word Edition; file:285 is OMNICELLULAR-AI-Collaboration-Bootstrap.md. These references establish document provenance; they do not imply independent confirmation of the original report's external literature.